



Roots of Polynomials · Mark Scheme

Core Pure 1 Chapter 4: Roots of Polynomials. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Sums and products of roots

- 1 $\alpha + \beta = \frac{5}{2}$ B1; $\alpha\beta = \frac{7}{2}$ B1.
- 2 $\sum \alpha = 2$ B1; $\sum \alpha\beta = \frac{4}{3}$ B1; $\alpha\beta\gamma = -3$ B1.
- 3 $\sum \alpha = 3$ B1; $\sum \alpha\beta = 2$ B1; $\sum \alpha\beta\gamma = 1$ and $\alpha\beta\gamma\delta = 5$ B1.
- 4 $6\alpha^3 = 48$ M1; $\alpha = 2$, roots 2, 4, 6 A1; $p = -(2+4+6)$ or $q = 8+24+12$ M1; $p = -12$, $q = 44$ A1.
- 5(a) $(a-d) + a + (a+d) = 3a = 9$ M1; $a = 3$ A1.
- 5(b) $3(3-d)(3+d) = 15$ M1; $9-d^2 = 5$, $d = \pm 2$ A1; roots 1 and 5 A1.
- 5(c) $k = \sum \alpha\beta = 1 \times 3 + 3 \times 5 + 5 \times 1$ M1; $k = 23$ A1.

Section B – Symmetric functions

- 6 $(\alpha + \beta)^2 - 2\alpha\beta = 16 - 14$ M1; $= 2$ A1.
- 7 (a) $\frac{\alpha + \beta}{\alpha\beta}$ M1; $= \frac{4}{7}$ A1. (b) $\frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{2}{7}$ A1.
- 8 $\sum \alpha = 2$, $\sum \alpha\beta = 3$ B1; $\sum \alpha^2 = 2^2 - 2(3) = -2$ M1; the sum of the squares of real numbers cannot be negative, so the roots are not all real A1.
- 9 $\sum \alpha = -\frac{1}{2}$, $\sum \alpha\beta = -2$, $\alpha\beta\gamma = -3$ B1; (a) $\frac{\sum \alpha\beta}{\alpha\beta\gamma}$ M1; $= \frac{2}{3}$ A1; (b) $\frac{\sum \alpha}{\alpha\beta\gamma} = \frac{1}{6}$ A1.
- 10(a) $\sum \alpha = -2$, $\sum \alpha\beta = -3$, $\alpha\beta\gamma = -5$ B1; $(-2)^2 - 2(-3)$ M1; $= 10$ A1.
- 10(b) Each root satisfies $\alpha^3 = -2\alpha^2 + 3\alpha - 5$, so $\sum \alpha^3 = -2\sum \alpha^2 + 3\sum \alpha - 15$ M1; $= -2(10) + 3(-2) - 15$ M1; $= -41$ A1.
Alternative: $\sum \alpha^3 = (\sum \alpha)^3 - 3\sum \alpha \sum \alpha\beta + 3\alpha\beta\gamma$.
- 10(c) $\alpha\beta\gamma + \sum \alpha\beta + \sum \alpha + 1$ M1; $= -5 - 3 - 2 + 1 = -9$ A1.

Section C – Transforming equations

- 11 Substitute $x = \frac{y}{2}$: $\frac{y^2}{4} - \frac{3y}{2} + 5 = 0$ M1; $y^2 - 6y + 20 = 0$ A1.
- 12 Substitute $x = y - 1$ M1; $(y-1)^3 - 4(y-1)^2 + (y-1) + 2 = 0$ expanded M1; $y^3 - 7y^2 + 12y - 4 = 0$ A1.
- 13 Substitute $x = \frac{1}{y}$: $\frac{2}{y^3} - \frac{1}{y^2} + \frac{3}{y} - 1 = 0$ M1; multiply by y^3 : $2 - y + 3y^2 - y^3 = 0$ M1; $y^3 - 3y^2 + y - 2 = 0$ A1.
- 14 $x = \sqrt{y}$: $y\sqrt{y} + 3y - 2\sqrt{y} + 1 = 0$ M1; $\sqrt{y}(y-2) = -(3y+1)$, square: $y(y-2)^2 = (3y+1)^2$ M1; $y^3 - 4y^2 + 4y = 9y^2 + 6y + 1$ M1; $y^3 - 13y^2 - 2y - 1 = 0$ A1.
- 15(a) $x = \frac{y+1}{2}$ B1; $\frac{2(y+1)^3}{8} + \frac{5(y+1)^2}{4} - \frac{3(y+1)}{2} + 4 = 0$ M1; $\times 4$: $(y+1)^3 + 5(y+1)^2 - 6(y+1) + 16 = 0$ M1; $y^3 + 8y^2 + 7y + 16 = 0$ A1.
- 15(b) (i) Product of roots $= -16$ B1; (ii) $(-8)^2 - 2(7)$ M1; $= 50$ A1.

Roar-level challenge

- 16 $\sum \alpha = -2$, $\sum \alpha\beta = -1$, $\alpha\beta\gamma = -3$; sum of new roots $= \sum \alpha\beta = -1$ B1; sum of products in pairs $= \alpha\beta\gamma(\alpha + \beta + \gamma)$ M1; $= (-3)(-2) = 6$ A1; product $= (\alpha\beta\gamma)^2 = 9$ B1; $y^3 + y^2 + 6y - 9 = 0$ A1.